

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I _____ (V. Asha sarthiwika / 192311066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:V. Asha Sarthiwika

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: SoilDry = True, CropWheat = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Annexure A – Case Study Solutions

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Let the following symbols represent the propositions:

- **F** = Patient A has Fever
- **C** = Patient A has Cough
- **B** = Patient A has Breathlessness
- **Flu** = Patient A has Possible Flu
- **M** = Patient A has Possible Measles
- **P** = Patient A has Possible Pneumonia

The rules are represented as:

1. **Fever AND Cough $\rightarrow$ Possible Flu**
2. **Fever AND Rash $\rightarrow$ Possible Measles**

If **R** = Patient A has Rash:

3. **Cough AND Breathlessness $\rightarrow$ Possible Pneumonia**

Facts:

There is no fact stating that the patient has a rash, so **R is not known to be true.**

(b) Forward Chaining

Forward chaining starts with the known facts and repeatedly applies rules whose conditions are satisfied.

Initial facts:

Step 1 – Apply Rule 1

Rule:

Since:

and

both conditions are satisfied.

Therefore:

is derived.

Step 2 – Apply Rule 2

Rule:

We know Fever is true, but **Rash is not given as a fact.**

Therefore, Rule 2 cannot be fired.

Step 3 – Apply Rule 3

Rule:

Both conditions are true:

Therefore:

is derived.

Final Forward-Chaining Conclusions

Possible Measles **cannot be derived** because there is no fact that Patient A has a rash.

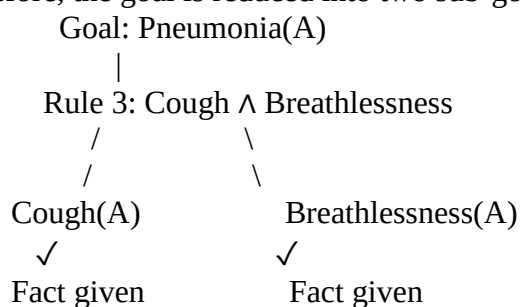
(c) Backward Chaining for Pneumonia

Goal:

Look for a rule whose conclusion is Pneumonia.

Rule 3:

Therefore, the goal is reduced into two sub-goals:



Both sub-goals are facts in the knowledge base.
Therefore:

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification of Rule 1

Rule 1 is:

Given facts:

We need to match:

with:

Therefore:

Similarly,

matches:

with the same substitution:

Applying the substitution to Rule 1:

Both facts are true, so:

is derived.

MGU

This is the most general unifier because the variable only needs to be replaced by the constant **Ambulance**.

(b) Resolution Proof of Proceed(CarA)

Relevant rules:

Rule 1

Convert implication:

Rule 2

This gives two clauses:

Rule 3

CNF:

Facts

Resolution Steps

From Rule 1:

Substitute :

Resolve with:

Result:

Resolve with:

Result:

Now use Rule 2:

Resolve with :

Now Rule 3:

Resolve with:

gives:

Resolve with:

gives:

Resolve with:

gives:

Resolution Refutation

To prove the goal formally, add its negation:

Resolving this with the derived clause:

gives:

But we already derived:

Therefore:

resolve to:

Hence:

(c) Handling Two Simultaneous Emergency Vehicles

The current knowledge base is **partially sufficient** for two emergency vehicles because Rule 1 is universally quantified:

Therefore, if we have:

and

the same rule can derive:

and

However, the current KB does not contain rules for **conflict resolution** when both emergency vehicles require the same road or intersection.

Additional facts/rules could include:

1. **Emergency priority**
2. **Conflict detection**
3. **Priority-based signal control**
4. **Hold the lower-priority vehicle**

Thus, the existing KB can independently process multiple emergency vehicles, but **additional rules are required to resolve simultaneous route conflicts safely**.

Case Study 3 — Agricultural Expert Reasoning System

(a) Conversion into CNF

P1

Original:

Step 1 – Eliminate implication

Using:

we get:

This is already CNF.

P2

Original:

Eliminate implication:

Apply De Morgan's law:

Therefore:

P3

Original:

Eliminate implication:

Using De Morgan's law:

Therefore:

Facts

Given:

and:

Their CNF representation is simply:

Complete CNF

(b) Resolution Refutation for ApplyDripMethod

Step 1 – Goal

Goal:

Negate the goal:

Add this to the KB.

Step 2 – Resolve P1 with SoilDry

P1:

Fact:

Resolution gives:

Step 3 – Resolve P2 with CropWheat

P2:

We have:

and:

Resolve with :

Resolve with :

Step 4 – Resolve with Negated Goal

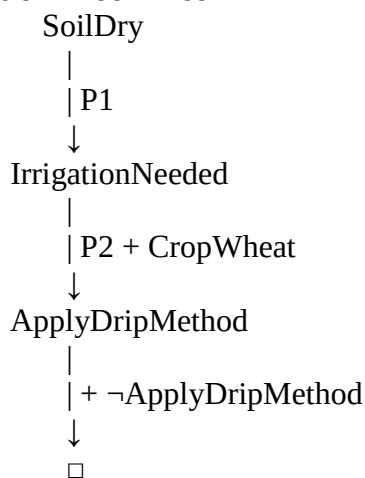
We added:

and derived:

Therefore:

resolve to:

Resolution Proof Tree



Therefore:

(c) Removing SoilDry

Suppose the fact:

is removed.

The remaining facts are:

P1 remains:

But we no longer know that SoilDry is true.

Therefore, we **cannot derive**:

Without IrrigationNeeded, P2 cannot derive:

So the reasoning stops there.

Does CropAtRisk follow?

P3 is:

We know:

But we do **not** know:

Not being able to prove ApplyDripMethod does **not** mean that its negation is true. This is an important principle of classical logic.

Therefore:

Final conclusion

Without SoilDry:

- IrrigationNeeded → **Cannot be derived**
- ApplyDripMethod → **Cannot be derived**
- ¬ApplyDripMethod → **Not given**
- CropAtRisk → **Cannot be derived**

Thus, the correct conclusion is:

Case Study 4 — Wumpus World Knowledge Agent

(a) Belief State and Modus Ponens

The agent receives the following percepts:

Therefore, the initial belief state is:

Rule 1

We know:

Therefore, by Modus Ponens:

Rule 2

We know:

Therefore:

Rule 3

Given:

Substitute:

Therefore:

Rule 4

There are no explicit facts giving:

or:

Therefore, Rule 4 cannot currently be fired.

Derived Belief State

The agent can derive:

It **cannot derive specific Wumpus/Pit locations or Safe locations** from the supplied rules alone.

(b) Forward Chaining for Possible Wumpus and Pit Locations

Step 1 – Stench

Given:

Rule 1 gives:

Therefore:

In a standard Wumpus World, a stench means that the Wumpus is in **one of the cells adjacent to [1,2]**.

Assuming the usual four-directional grid adjacency, the candidate cells are:

Thus:

assuming all these cells exist in the world.

Step 2 – Breeze

Given:

Rule 2 gives:

Therefore:

Under standard four-directional adjacency, the neighboring cells are:

Therefore:

assuming those cells exist.

Step 3 – Glitter

Given:

Rule 3 gives:

Important Logical Limitation

The given rules tell us that the Wumpus/Pit are **adjacent**, but they do not provide enough information to identify one unique location.

Therefore, the logical result is a **set of possible locations**, not one exact location.

(c) Safety of Path [1,1] → [1,2] → [2,2]

The proposed path is:

Cell [1,1]

We have:

This indicates that a pit may be adjacent to [1,1]. It does **not** prove that [1,1] itself contains a pit. However, the supplied KB does not explicitly establish:

Therefore, according to the given logical rules, [1,1] cannot be proven safe.

Cell [1,2]

We have:

Therefore, a Wumpus is adjacent to [1,2].

But the rules do not prove that the Wumpus is actually in [1,2].

Also, there is no explicit:

Therefore, [1,2] cannot be proven safe.

Cell [2,2]

We know:

Therefore:

The agent wants to collect the gold here. However, the given KB does not establish:
and

Therefore, Rule 4 cannot establish:

Final Safety Conclusion

The path:

cannot be logically proven safe from the given knowledge base.

The agent knows that:

- [1,2] has a **stench**, indicating a Wumpus in an adjacent cell.
- [1,1] has a **breeze**, indicating a pit in an adjacent cell.
- [2,2] contains **gold**.
- No explicit negative facts establish that the path cells are free of Wumpus or pits.

Hence:

This is different from saying that the path is definitely dangerous. **The correct logical conclusion is that safety is not proven with the available information.**